

Lab 1

Object Explorer | Dashboard | dvdrntal/postgres@PostgreSQL 16* | dvdrntal/postgre...

PostgreSQL 16

- Databases (2)
 - dvdrntal
 - Cast
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers
 - Languages
 - Publications
 - Schemas (1)
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (15)
 - Trigger Functions
 - Types
 - Views
 - postgres
 - Cast
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers

Query | Query History | Scratch Pad

```
1 BEGIN;
```

Data Output | Messages | Notifications

BEGIN

Query returned successfully in 45 msec.

Total rows: Query complete 00:00:00.045 | LF | Ln 1, Col 7

Object Explorer | Dashboard | dvdrntal/postgre... | dvdrntal/postgres@PostgreSQL 16* |

PostgreSQL 16

- Databases (2)
 - dvdrntal
 - Cast
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers
 - Languages
 - Publications
 - Schemas (1)
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (15)
 - Trigger Functions
 - Types
 - Views
 - postgres
 - Cast
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers

Query | Query History | Scratch Pad

```
1 UPDATE film SET rental_rate = 5.99 WHERE film_id = 1;
```

Data Output | Messages | Notifications

Total rows: Waiting for the query to complete... 00:00:04.038 | LF | Ln 1, Col 54

The image displays two sequential screenshots of the pgAdmin 4 interface, illustrating a database transaction. The left sidebar shows the 'Object Explorer' with the 'dvdrental' database selected. The main window shows the 'Query' editor with the following SQL statement:

```
1 UPDATE film SET rental_rate = 5.99 WHERE film_id = 1;
```

In the first screenshot, the query is executed, and the 'Data Output' pane shows the message: 'UPDATE 1' and 'Query returned successfully in 42 secs 621 msec.' The status bar at the bottom indicates 'Total rows: Query complete 00:00:42.621'.

In the second screenshot, the same query is executed again, followed by a 'COMMIT' statement:

```
1 UPDATE film SET rental_rate = 5.99 WHERE film_id = 1;  
2 COMMIT;
```

The 'Data Output' pane shows the message: 'COMMIT' and 'Query returned successfully in 39 msec.' The status bar at the bottom indicates 'Total rows: Query complete 00:00:00.039'. A green notification box at the bottom right of the second screenshot states: '✓ Query returned successfully in 39 msec. ✕'.

Khi Session A mở transaction và UPDATE row `film_id = 1`, PostgreSQL cấp cho A một Exclusive Lock (X-Lock) trên row đó. Khi Session B cố UPDATE cùng row, B phải xin cũng X-Lock nhưng theo ma trận tương thích lock, X-Lock không tương thích với X-Lock đang tồn tại => B bị block. Thời gian chờ của B (42.6 giây) chính bằng khoảng thời gian A giữ transaction mở trước khi COMMIT. Ngay khi A COMMIT, lock được giải phóng, B lập tức được cấp lock và UPDATE thành công.

Lab 2

The screenshot displays the pgAdmin interface for a PostgreSQL database named 'dvdrental'. The left sidebar shows the 'Object Explorer' with the 'public' schema expanded, listing various database objects. The main pane shows a SQL query: `SELECT rental_rate FROM film WHERE film_id = 10;`. The 'Data Output' pane at the bottom shows the result of the query: a single row with 'rentalRate' as 'numeric(4,2)' and a value of '0.99'. The status bar at the bottom indicates 'Total rows: 1' and 'Query complete 00:00:00.078'.

Session A mở transaction và SELECT rental_rate lần 1, ra kết quả X. Trong lúc A vẫn đang mở (chưa COMMIT), Session B UPDATE cùng row đó thành giá trị mới và COMMIT thành công vì ở mức cô lập mặc định Read Committed, S-Lock của A chỉ được giữ trong lúc câu SELECT chạy, không giữ xuyên suốt cả transaction, nên B không hề bị B chặn. Khi A SELECT lại lần 2 (vẫn trong cùng transaction, chưa COMMIT), Postgres cho A đọc bản ghi đã commit mới nhất => ra giá trị khác lần đầu.

Lab 3

Session A giữ X-lock trên row film_id=100 và cố xin thêm X-lock trên row film_id=200. Đồng thời Session B giữ X-lock trên row film_id=200 và cố xin thêm X-lock trên row film_id=100. Hai transaction tạo thành một Wait-for Graph có chu trình khép kín: A chờ B chờ A. PostgreSQL không ngăn chặn deadlock trước (vì tốn chi phí), mà dùng cơ chế Deadlock Detection: sau khoảng thời gian deadlock_timeout (mặc định khoảng 1 giây), Postgres quét đồ thị chờ, phát hiện chu trình, và tự động chọn 1 victim (ở đây là Session B) để ROLLBACK báo lỗi ERROR: deadlock detected (SQL state 40P01). Transaction còn lại (Session A) được giải phóng lock và tiếp tục chạy bình thường tới khi COMMIT.

Object Explorer PostgreSQL 16

- Databases (2)
 - dvdrental
 - Casts
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers
 - Languages
 - Publications
 - Schemas (1)
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (15)
 - Trigger Functions
 - Types
 - Views
 - postgres
 - Casts
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers

Dashboard X dvdrental/postgres@PostgreSQL 16* X dvdrental/postgre...

dvdrental/postgres@PostgreSQL 16

Query Query History Scratch Pad X

```
1 BEGIN;
2 UPDATE film SET rental_rate = 2.99 WHERE film_id = 100;
```

Data Output Messages Notifications

WARNING: there is already a transaction in progress
UPDATE 1

Query returned successfully in 46 msec.

Total rows: Query complete 00:00:00.046 LF Ln 2, Col 56

Object Explorer PostgreSQL 16

- Databases (2)
 - dvdrental
 - Casts
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers
 - Languages
 - Publications
 - Schemas (1)
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (15)
 - Trigger Functions
 - Types
 - Views
 - postgres
 - Casts
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers

Dashboard X dvdrental/postgre... X dvdrental/postgres@PostgreSQL 16* X

dvdrental/postgres@PostgreSQL 16

Query Query History Scratch Pad X

```
1 BEGIN;
2 UPDATE film SET rental_rate = 2.99 WHERE film_id = 200;
```

Data Output Messages Notifications

WARNING: there is already a transaction in progress
UPDATE 1

Query returned successfully in 49 msec.

Total rows: Query complete 00:00:00.049

✓ Query returned successfully in 49 msec.

LF Ln 2, Col 56

Object Explorer PostgreSQL 16

- Databases (2)
 - dvdrental
 - Casts
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers
 - Languages
 - Publications
 - Schemas (1)
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (15)
 - Trigger Functions
 - Types
 - Views
 - Subscriptions
 - postgres
 - Casts
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers

Dashboard X dvdrental/postgres@PostgreSQL 16* X dvdrental/postgre... X

dvdrental/postgres@PostgreSQL 16

Query Query History Scratch Pad X

```

1 BEGIN;
2 UPDATE film SET rental_rate = 2.99 WHERE film_id = 100;
3 UPDATE film SET rental_rate = 3.99 WHERE film_id = 200;

```

Data Output Messages Notifications

WARNING: there is already a transaction in progress

Total rows: Waiting for the query to complete... 00:00:29.037 LF Ln 3, Col 56

Object Explorer PostgreSQL 16

- Databases (2)
 - dvdrental
 - Casts
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers
 - Languages
 - Publications
 - Schemas (1)
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (15)
 - Trigger Functions
 - Types
 - Views
 - Subscriptions
 - postgres
 - Casts
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers

Dashboard X dvdrental/postgre... X dvdrental/postgres@PostgreSQL 16* X

dvdrental/postgres@PostgreSQL 16

Query Query History Scratch Pad X

```

1 BEGIN;
2 UPDATE film SET rental_rate = 2.99 WHERE film_id = 200;
3 UPDATE film SET rental_rate = 3.99 WHERE film_id = 100;

```

Data Output Messages Notifications

WARNING: there is already a transaction in progress
 ERROR: deadlock detected
 Process 96796 waits for ShareLock on transaction 966; blocked by process 96789.
 Process 96789 waits for ShareLock on transaction 967; blocked by process 96796.
 SQL state: 40P01
 Detail: Process 96796 waits for ShareLock on transaction 966; blocked by process 96789.
 Process 96789 waits for ShareLock on transaction 967; blocked by process 96796.
 Hint: See server log for query details.
 Context: while locking tuple (5,12) in relation "film"

Total rows: Query complete 00:00:01.112 LF Ln 3, Col 56

Lab 4

Object Explorer PostgreSQL 16

- Databases (2)
 - dvdrental
 - Casts
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers
 - Languages
 - Publications
 - Schemas (1)
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (15)
 - Trigger Functions
 - Types
 - Views
 - postgres
 - Casts
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers

Dashboard X dvdrental/postgres@PostgreSQL 16* X dvdrental/postgre...

dvdrental/postgres@PostgreSQL 16

Query Query History

```
1 BEGIN;  
2 SELECT * FROM inventory WHERE inventory_id = 10 FOR UPDATE;
```

Scratch Pad X

Data Output Messages Notifications

inventory_id [PK] integer	film_id smallint	store_id smallint	last_update timestamp without time zone
1	10	2	2006-02-15 10:09:17

Showing rows: 1 to 1 Page No: 1 of 1

Total rows: 1 Query complete 00:00:00.061

Successfully run. Total query runtime: 61 msec. 1 rows affected.

Object Explorer PostgreSQL 16

- Databases (2)
 - dvdrental
 - Casts
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers
 - Languages
 - Publications
 - Schemas (1)
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (15)
 - Trigger Functions
 - Types
 - Views
 - postgres
 - Casts
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers

Dashboard X dvdrental/postgre... X dvdrental/postgres@PostgreSQL 16* X

dvdrental/postgres@PostgreSQL 16

Query Query History

```
1 BEGIN;  
2 SELECT * FROM inventory WHERE inventory_id = 10 FOR UPDATE;
```

Scratch Pad X

Data Output Messages Notifications

WARNING: there is already a transaction in progress

Total rows: Waiting for the query to complete... 00:00:12.235

Object Explorer: PostgreSQL 16, Databases (2), dvdrental, public, Schemas (1), public, Aggregates, Collations, Domains, FTS Configurations, FTS Dictionaries, FTS Parsers, FTS Templates, Foreign Tables, Functions, Materialized Views, Operators, Procedures, Sequences, Tables (15), Trigger Functions, Types, Views, Subscriptions, postgres, Casts, Catalogs, Event Triggers, Extensions, Foreign Data Wrappers.

Query: Query History

```

1 BEGIN;
2 SELECT * FROM inventory WHERE inventory_id = 10 FOR UPDATE;
3 INSERT INTO rental (rental_date, inventory_id, customer_id, staff_id)
4 VALUES (NOW(), 10, 1, 1);

```

Data Output: Messages, Notifications

WARNING: there is already a transaction in progress
INSERT 0 1

Query returned successfully in 70 msec.

Total rows: Query complete 00:00:00.070

Object Explorer: PostgreSQL 16, Databases (2), dvdrental, public, Schemas (1), public, Aggregates, Collations, Domains, FTS Configurations, FTS Dictionaries, FTS Parsers, FTS Templates, Foreign Tables, Functions, Materialized Views, Operators, Procedures, Sequences, Tables (15), Trigger Functions, Types, Views, Subscriptions, postgres, Casts, Catalogs, Event Triggers, Extensions, Foreign Data Wrappers.

Query: Query History

```

1 BEGIN;
2 SELECT * FROM inventory WHERE inventory_id = 10 FOR UPDATE;

```

Data Output: Messages, Notifications

inventory_id	film_id	store_id	last_update
10	2	2	2006-02-15 10:09:17

Total rows: 1 Query complete 00:00:42.758

SELECT ... FOR UPDATE cho phép ứng dụng chủ động xin X-lock ngay từ bước đọc, thay vì chỉ xin S-lock như SELECT thường. Khi Session A chạy câu này trên `inventory_id = 10`, nó giữ X-lock trên row đó suốt cho tới khi COMMIT/ROLLBACK. Khi Session B (đại diện khách hàng 2) cũng cố SELECT FOR UPDATE trên cùng row, B bị block ngay từ bước đọc khác hẳn SELECT thường (Read Committed) không bao giờ bị chặn bởi SELECT khác. B phải chờ tới khi A hoàn tất INSERT + COMMIT thì mới được cấp lock và đọc được dữ liệu. Nhờ đó,

ứng dụng ở B có thể kiểm tra logic (ví dụ: check xem DVD đã có rental record chưa) một cách an toàn, không có race condition, tránh được tình huống 2 khách hàng cùng thuê 1 bản DVD duy nhất.